
CHEMISTRY**0620/62**

Paper 6 Alternative to Practical

March 2019

MARK SCHEME

Maximum Mark: 40

Published

This mark scheme is published as an aid to teachers and candidates, to indicate the requirements of the examination. It shows the basis on which Examiners were instructed to award marks. It does not indicate the details of the discussions that took place at an Examiners' meeting before marking began, which would have considered the acceptability of alternative answers.

Mark schemes should be read in conjunction with the question paper and the Principal Examiner Report for Teachers.

Cambridge International will not enter into discussions about these mark schemes.

Cambridge International is publishing the mark schemes for the March 2019 series for most Cambridge IGCSE™, Cambridge International A and AS Level components and some Cambridge O Level components.

PUBLISHED**Generic Marking Principles**

These general marking principles must be applied by all examiners when marking candidate answers. They should be applied alongside the specific content of the mark scheme or generic level descriptors for a question. Each question paper and mark scheme will also comply with these marking principles.

GENERIC MARKING PRINCIPLE 1:

Marks must be awarded in line with:

- the specific content of the mark scheme or the generic level descriptors for the question
- the specific skills defined in the mark scheme or in the generic level descriptors for the question
- the standard of response required by a candidate as exemplified by the standardisation scripts.

GENERIC MARKING PRINCIPLE 2:

Marks awarded are always **whole marks** (not half marks, or other fractions).

GENERIC MARKING PRINCIPLE 3:

Marks must be awarded **positively**:

- marks are awarded for correct/valid answers, as defined in the mark scheme. However, credit is given for valid answers which go beyond the scope of the syllabus and mark scheme, referring to your Team Leader as appropriate
- marks are awarded when candidates clearly demonstrate what they know and can do
- marks are not deducted for errors
- marks are not deducted for omissions
- answers should only be judged on the quality of spelling, punctuation and grammar when these features are specifically assessed by the question as indicated by the mark scheme. The meaning, however, should be unambiguous.

GENERIC MARKING PRINCIPLE 4:

Rules must be applied consistently e.g. in situations where candidates have not followed instructions or in the application of generic level descriptors.

GENERIC MARKING PRINCIPLE 5:

Marks should be awarded using the full range of marks defined in the mark scheme for the question (however; the use of the full mark range may be limited according to the quality of the candidate responses seen).

GENERIC MARKING PRINCIPLE 6:

Marks awarded are based solely on the requirements as defined in the mark scheme. Marks should not be awarded with grade thresholds or grade descriptors in mind.

Question	Answer	Marks
1(a)	Electrolysis	1
1(b)	Open (air hole / collar)	1
1(c)	Fizzing / bubbles / green gas	1
	Chlorine / Cl_2	1
1(d)	Iron reacts (with <u>chlorine</u>) / iron is reactive / not inert	1
1(e)(i)	Bubbles / effervescence / fizzing	1
	Hydrogen (is below zinc in the reactivity series)	1
1(e)(ii)	Litmus	1
	Bleached / turns white	1
1(f)	Wear gloves / goggles	1

Question	Answer	Marks
2(a)	Table of results for Experiment 1	1
	Initial and final volumes completed correctly 1.1, 16.1	
	Difference 15.0	1
2(b)	Table of results for Experiment 2	1
	Initial, final volumes and difference completed correctly 1.9, 31.9, 30.0	
	All readings to 1 dp in (a) and (b)	1

Question	Answer	Marks
2(c)(i)	Solution A	1
	Smaller volume / less (of potassium manganate used) / solution A	1
2(c)(ii)	2 · (times) as concentrated	1
2(d)(i)	2 · (times) value from table for Experiment 2, 60 cm ³	1
	Double volume (of C) used	1
2(d)(ii)	Volume of potassium manganate solution added > 50 cm ³	1
	Use more than one burette / refill burette	1
2(e)	Advantage easy to use / quick	1
	Disadvantage not as accurate owtte	1
2(f)	(Solution C) contains iron(II) (ions) / Fe ²⁺	1
	Reference to oxidation / (iron(II)) → iron(III) / Fe ³⁺	1

Question	Answer	Marks
Tests on solution D		
3(a)	1–3	1
3(b)	Bubbles / fizz / effervescence	1
	Lighted splint / flame	1
	'Pops'	1

Question	Answer	Marks
3(c)	No reaction / no change / no precipitate	1
3(d)	White precipitate	1
Tests on solid E		
3(e)	Carbon dioxide	1
3(f)	Calcium	1
	Carbonate	1

Question	Answer	Marks
4	6 from: Weighed amount / x gram of magnesium Add known volume of dilute hydrochloric acid gas syringe / measuring cylinder over water Use of stop-clock / timer Measure volume of hydrogen at fixed time or time for a fixed volume to be made Repeat using different temperatures Compare results / conclusion	6